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(54) **GEODESIC DOME USING PRE-CASTED
PANELS OF MAGNESIUM-PHOSPHATE
CERAMIC CEMENT CONNECTED WITH A
PLURALITY OF HUBS.**

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(57) **ABSTRACT**

A Geodesic Dome made of triangular panels pre-casted with Magnesium Phosphate ceramic cement is described. Said triangular panels consist of multiple layers of Magnesium Phosphate ceramic cement poured into a mold with successive layers at various levels of porosity, reinforced with basalt and hemp fibers. A plurality of hubs joins a plurality of triangular panels at specific dihedral angles to enclose space. The hubs are bolted into a threaded sleeve that is embedded into the corner of each triangular panel, thus creating a Geodesic Dome.

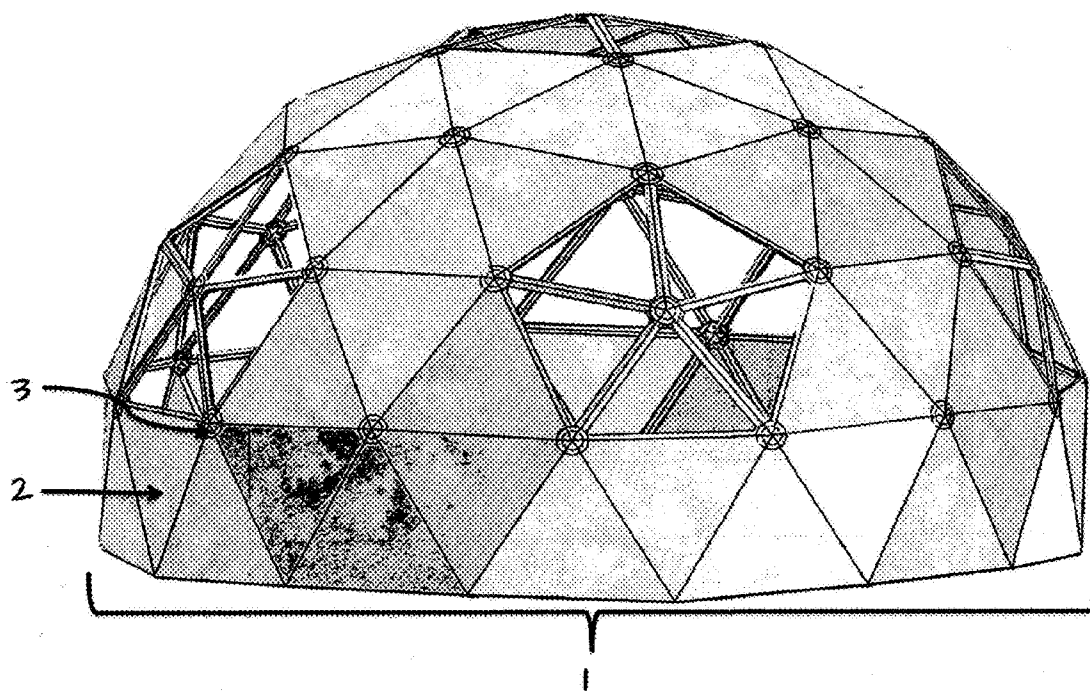


Figure #1

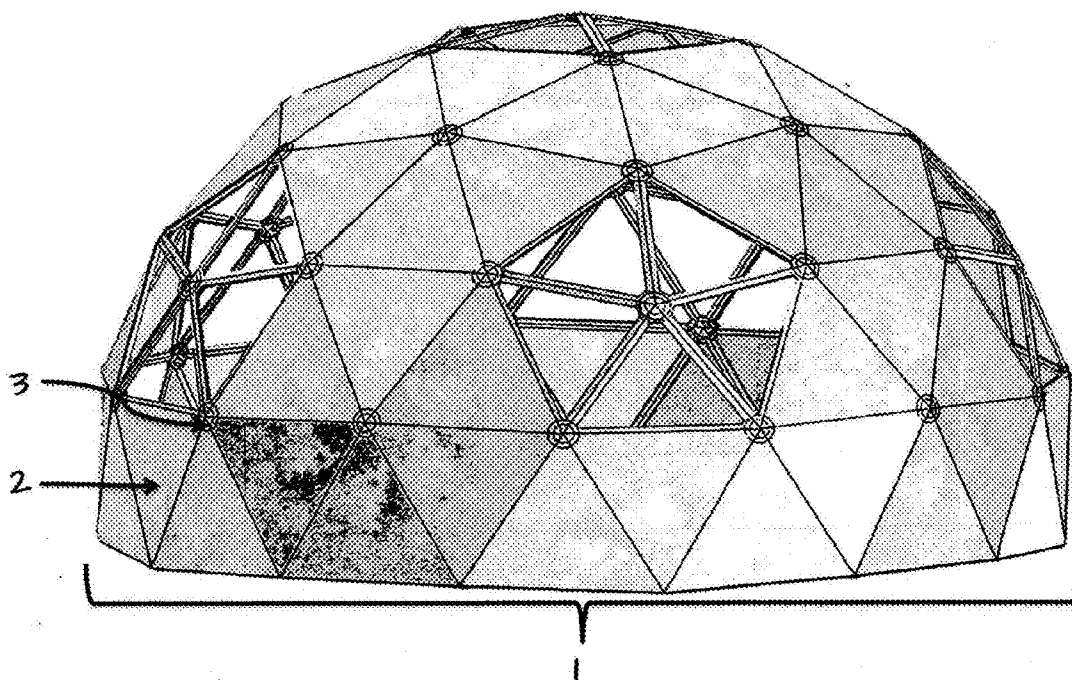


Figure #2

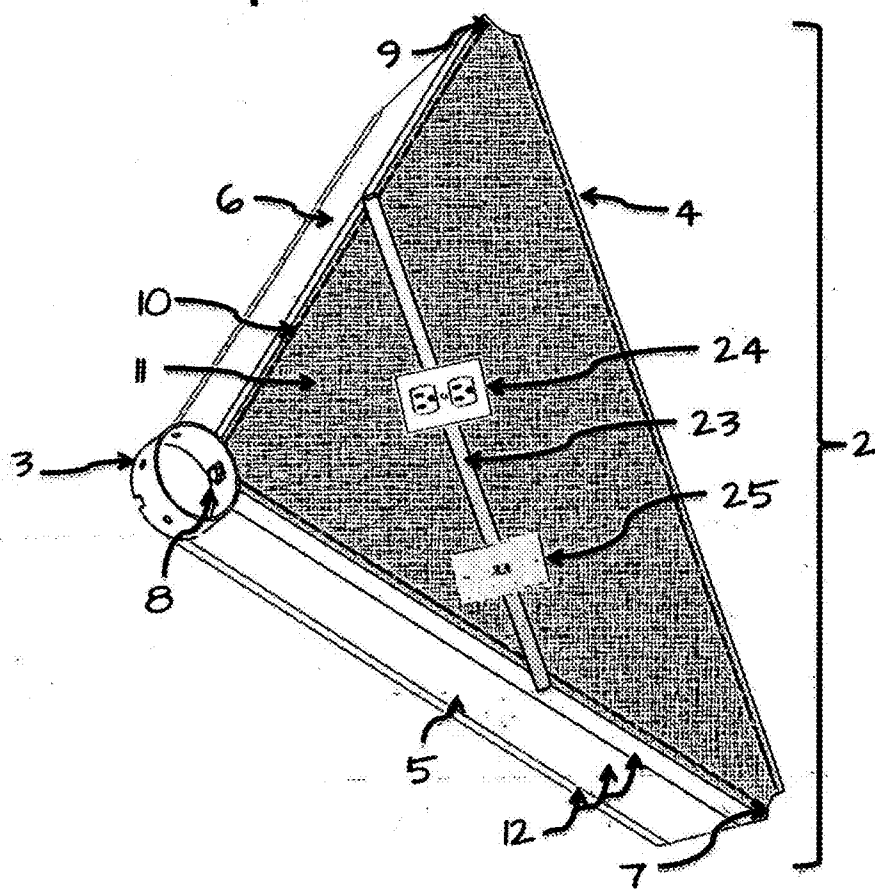


Figure #3

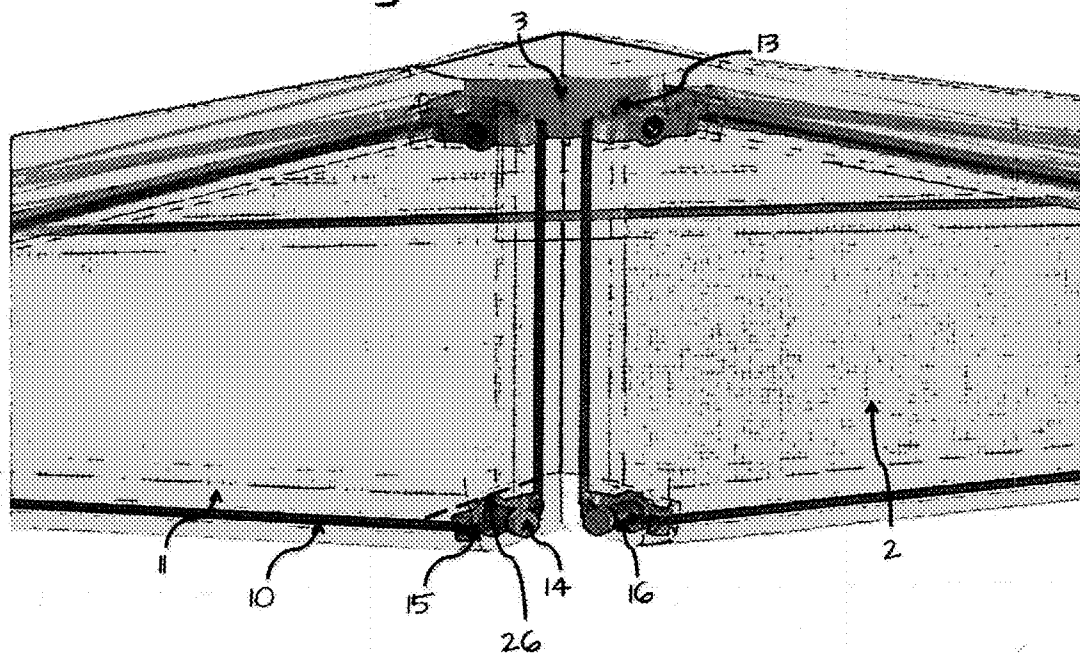
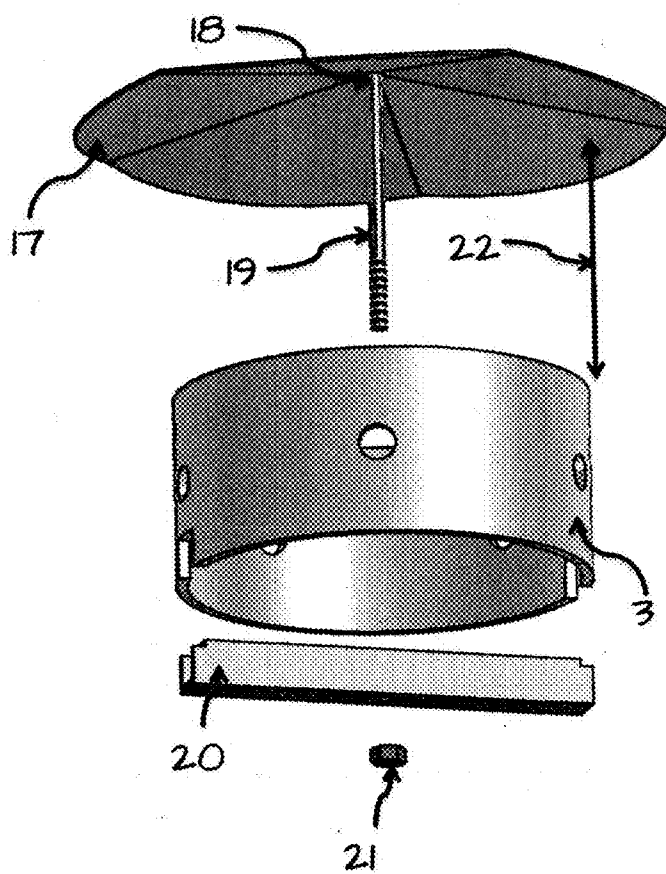


Figure #4



**GEODESIC DOME USING PRE-CASTED
PANELS OF MAGNESIUM-PHOSPHATE
CERAMIC CEMENT CONNECTED WITH A
PLURALITY OF HUBS.**

FIELD OF THE INVENTION

[0001] This invention relates to the manufacture of a Geodesic Dome to serve as shelter. The present invention utilizes triangular panels pre-casted of Magnesium Phosphate ceramic cement reinforced with Basalt Fibers and connected by a plurality of hubs. The result is a pleasant environment for human occupation.

BACKGROUND OF THE INVENTION

[0002] The Geodesic Dome is well recognized as the optimal method of enclosing the maximum space with the minimum amount of materials. The fundamental geometry of this structure creates many challenges when traditional construction methods are applied. Traditional construction using lumber, plywood, and shingles is complex, creates excessive waste, is difficult to maintain, and difficult to make waterproof. The introduction of Magnesium Phosphate ceramic cement enables pre-casting of extremely strong and waterproof triangular panels, which are capable of withstanding centuries of deterioration. Most of the pitfalls of traditional Geodesic Dome construction have been overcome with the present invention.

SUMMARY OF THE INVENTION

[0003] A method is described for constructing a Geodesic Dome using precast triangular panels made from Magnesium Phosphate ceramic cement reinforced with Basalt Fibers. Multiple layers of Magnesium Phosphate ceramic cement at various densities are poured into a mold forming the triangular panel.

[0004] A plurality of hubs join a plurality of triangular panels at specific dihedral angles to create the Geodesic Dome. Each hub has five or six holes for a bolt to be tightened into a threaded sleeve which is embedded into the corners of each triangular panel. Connecting the triangular panels together with hubs using bolts makes for a static connection. The hub is fitted with a cover to seal the open void. The cover is shaped as a domed pentagon or domed hexagon, with an apex hole for a threaded rod. The threaded rod is used to tighten the cover, creating a waterproof seal. Gaskets are utilized between the seams of the triangular panels to create a water-proof seal. The end result is a fully enclosed and waterproof Geodesic Dome.

[0005] An alternative version of the triangular panels is offered as skylights. The covers may be opened for ventilation. Decorative textures, aggregates, and colorings can be added to the exterior or interior surface of the triangular panel. An additional thin coat of specially formulated ceramic cement can be added to shield the environment from undesirable frequencies.

[0006] The net result is a Geodesic Dome that is visually appealing, easy to assemble, easy to disassemble for transport, strong, and long-lasting. The resulting Geodesic Dome is capable of withstanding severe weather conditions, generations of aging, and is manufactured without ecological offense. The resulting Geodesic Dome has excellent thermal qualities and makes for a pleasant enclosed space.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The invention will be described with reference to the accompanying drawings, in which component parts are labeled 1-21.

[0008] FIG. 1: Illustrates a high level view of the invented device. Shown is a plurality of triangular panels interconnected into a Geodesic Dome.

[0009] FIG. 2: illustrates a typical triangular panel of the invented device,

[0010] FIG. 3: Illustrates an assembly of five triangular panels forming a domed pentagon,

[0011] FIG. 4: Illustrates a typical hub and cap assembly of the invented device.

DETAILED DESCRIPTION OF THE INVENTION

[0012] FIG. 1 illustrates a high level view of the invented device. Shown is a Geodesic Dome (1) with a plurality of triangular panels (2). A plurality of hubs (3) join said plurality of triangular panels (2) to form said Geodesic Dome (1). Said triangular panels (2) are set at specific dihedral angles to create said Geodesic Dome (1). A plurality of gaskets is used to create waterproof seams around each triangular panel (2).

[0013] FIG. 2 illustrates a typical triangular panel (2) of the invented device. Multiple layers (12) of Magnesium Phosphate Ceramic Cement are poured into a mold at various levels of density. This makes for a structurally strong and well-insulated Geodesic Dome (1). Said triangular panel (2) is bounded by first edge face (4), second edge face (5), and third edge face (6). First edge face (4), second edge face (5), and third edge face (6) are sloped to create dihedral angles of the Geodesic Dome (1). Said triangular panel (2) contains first hub receiver (7), second hub receiver (8), and third hub receiver (9) to receive hub (3). An Integral matrix of basalt fibers, including basalt rebar (10), and basalt mesh (11) serve to reinforce said triangular panel (2). Multiple layers (12) of Magnesium Phosphate Ceramic cement at various levels of density create a strong and insulated triangular panel (2).

[0014] FIG. 3 illustrates five triangular panels (2) joined with a hub (3) to create a domed pentagon shape. Alternatively, six triangular panels (2) are joined with a hub (3) to create a domed hexagon shape. Said hub (3) contains 5-6 bolt holes (13) for 5-6 bolts (14) to be utilized in connecting said hub (3) to said triangular panels (2). Corner hardware (15) including a threaded sleeve (16) is embedded into corner of each triangular panel (2). Said corner hardware (15) and said threaded sleeve (16) are utilized to form an integral assembly of basalt rebar (10) and basalt mesh (11), which is bolted to said hub (3).

[0015] FIG. 4 shows a typical cover (17) for a typical hub (3). The cover is shaped as a domed pentagon or a domed hexagon, creating an apex (18). A threaded rod (19) connects said apex (18) to a mounting (20). Said cover (18) is tightened with a nut (21) against a gasket to create a waterproof seal around said hub (3).

[0016] The net result is a Geodesic Dome (1) that is strong, well insulated, easy to assemble, easy to transport, environmentally friendly, and resistant to deterioration, making for a very efficient human shelter. The pre-casting of triangular panels (2) using multi-layered Magnesium-Phosphate ceramic cement with Basalt reinforcement, joined together with a plurality of hubs (3), and fitted with a plurality of covers (18) is new and novel. Thus, leading to an advancement of art in Geodesic Dome (1) construction.

1. (canceled)
2. A geodesic dome comprising:
a plurality of panels; wherein
said plurality of panels are comprising magnesium phosphate ceramic cement; and
each of said plurality of panels is connected to at least one other of said plurality of panels.
3. The geodesic dome of claim 2; wherein
said plurality of panels are connected using a plurality of hubs; wherein
said plurality of hubs are polygonal in shape; and
said plurality of hubs comprising a plurality of holes for receiving bolts; and
each of said plurality of hubs having an open void; wherein
said open void is sealed with a cover.
4. The geodesic dome of claim 3; wherein
said cover is shaped as a domed polygon with apex; wherein
said apex is attached to a threaded rod; and
said threaded rod opposite end attached to a mounting brace in said hub; and
said cover tightens against at least one of said plurality of panels creating a waterproof seal.
5. The geodesic dome of claim 3; wherein
each of said plurality of panels comprise three corners; wherein
each of said three corners feature a ridge; wherein
said cover tightens against a ridge on at least one of said plurality of panels for an improved seal.
6. The geodesic dome of claim 3; wherein
said cover can be opened for ventilation.
7. The geodesic dome of claim 2; wherein
said plurality of panels are comprised of magnesium phosphate ceramic cement and an aggregate material; wherein
said aggregate material is selected from the group consisting of: quartz sand, silica sand, basalt fibers, hemp fibers; and an aqueous foam.
8. The geodesic dome of claim 7; wherein
said plurality of panels are comprising multiple layers of magnesium phosphate ceramic concrete at varying levels of density for strength and insulation.
9. The geodesic dome of claim 2; wherein
each one of said plurality of panels comprising at least three corners; wherein
each of said corners comprising corner hardware; wherein
said corner hardware connects an integral matrix of rebar and mesh; and
said corner hardware connects to a hub.
10. The geodesic dome of claim 9; wherein
said rebar and said mesh comprises materials selected from the group consisting of: basalt fibers, hemp fibers.
11. The geodesic dome of claim 2; wherein
at least one of said plurality of panels comprising an opening for a window or skylight.
12. The geodesic dome of claim 2; wherein
at least one of said plurality of panels comprising decorative textures, aggregates, and colorings for enhanced aesthetic appeal.
13. The geodesic dome of claim 2; wherein
at least one of said plurality of panels comprising conduit for electrical wires.
14. The geodesic dome of claim 2; wherein
at least one of said plurality of panels comprising electrical outlets and light switches.
15. The geodesic dome of claim 2; wherein
each of said plurality of panels is bounded by a first edge face, a second edge face, and a third edge face; wherein
said first edge face, said second edge face, and said third edge face being sloped to form the dihedral angles of a geodesic dome; and
said first edge face, said second edge face, and said third edge face being fitted with a gasket for a waterproof seal; and
each of said plurality of panels having three corners; a first hub receiver, a second hub receiver, a third hub receiver; wherein
said first hub receiver, said second hub receiver, and said third hub receiver having flat face for receiving polygonal hub; and
said first hub receiver, said second hub receiver, and said third hub receiver containing embedded corner hardware.
16. The geodesic dome of claim 15; wherein
said corner hardware is comprising a threaded sleeve.
17. The geodesic dome of claim 15; wherein
said first hub receiver, said second hub receiver, and said third hub receiver having concave curve for receiving circular hub.
18. The geodesic dome of claim 2; wherein
at least one of said plurality of triangular panels are modified to fit a door frame.
19. The geodesic dome of claim 2; wherein
at least one of said plurality of triangular panels comprise photovoltaic cells to generate electricity.

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